

Adversarial Co-Evolution as a Measurement Framework for Sandbagging Detection

Anonymous authors

Paper under double-blind review

Abstract

Capability evaluations are the primary safety signal for deployment. Sandbagging, which occurs when a model hides a capability during evaluation and reveals it in deployment, corrupts these signals. Existing detection work tests detectors against fixed attacks, but none measure what happens when the attacker adapts. Text-based detection turns out to be cheap to break: a two-line training-data edit erases the audit signal while leaving the attack intact, whereas representation-level detection is materially harder to evade. To show this, we built two attacks (xdomain and trigger) as small LoRA adapters on Qwen 2.5. We then ran a three-round red/blue co-evolution loop to trace exactly where each detector breaks, and evaluated four detection pillars across 8 model checkpoints and 4 model families. The text-only distributional audit collapsed from AUC 0.978 to 0.560 (chance level) after two training-data edits. On the other hand, the cross-model activation probe held $\text{AUC} \geq 0.979$ across all 8 models and fired regardless of trigger state. The key takeaway is that sandbagging detection should be evaluated by both accuracy and how much it costs an attacker to evade. Since no single method generalizes across all model families, multi-pillar coverage is necessary. The cost asymmetry is the deployable finding: text-based detection falls to a two-line edit, while activation-based detection requires a fundamental change to the fine-tuning signature.

1 Introduction

As defined by van der Weij et al. (2024), sandbagging is strategic underperformance on a capability evaluation. In our experiments, we define an attack as a concrete mechanism that causes sandbagging. We instantiate this as a LoRA adapter fine-tuned to suppress a target capability, either under a trigger phrase (trigger attack) or on a task format (xdomain attack). A developer fine-tunes a model to underperform on a targeted benchmark, either via a trigger phrase marking evaluation contexts or via format-selective training that causes failure on one task type while preserving performance on adjacent types. The blue team’s goal is to detect this without prior knowledge of the attack implementation. Prior detection work has converged on a shared assumption: that sandbagging is a problem with a single solution — find the right signal and flag it. We argue this framing is insufficient. Early approaches focused on behavioral elicitation. Tice et al. (2024) showed that injecting small amounts of Gaussian noise into model weights can expose suppressed capabilities in both prompted and password-locked sandbagging models. Working in activation space, Mack and Turner (2024) developed unsupervised steering-vector methods to mechanistically elicit latent behaviours. The common thread across this work is that each method was evaluated against a static or weakly adaptive attacker. We close this gap by introducing the adversarial co-evolution framework. The red team builds a LoRA-based attack and adapts it against whichever detector pillar currently produces the strongest signal; the blue team runs the full suite (logit, probe, behavioral, elicitation, and audit), measuring the break point at each iteration. The result is not a single pass/fail verdict on a detection method but a cost map of the attack–defense frontier, quantifying how much adversarial effort each pillar can absorb before failing (Figure 1).

We make the following contributions:

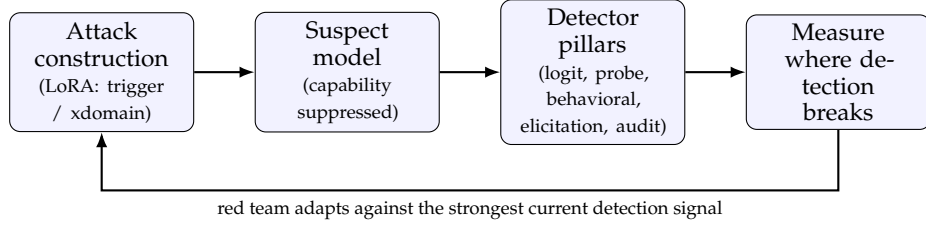


Figure 1: The adversarial co-evolution measurement framework. The red team builds an attack (a small LoRA adapter that suppresses a target capability), yielding a suspect model while the blue team runs detector pillars. We then measure where each detector breaks. The red team adapts against the strongest current signal and the loop repeats, quantifying the attacker cost to evade each detector (§3).

- **A format-selective attack that survives a cross-domain audit.** Our cross-domain (xdomain) attack produces 52–70 percentage-point GSM8K drops while preserving MMLU and arithmetic within $\pm 5\text{pp}$. Rather than removing the trigger, it confines the trigger’s effect to the targeted format, so an auditor running a standard cross-domain benchmark sweep—including math-adjacent subjects—would miss this attack entirely (§6).
- **A complete co-evolution arc with a documented endpoint.** We trace the xdomain attack’s distributional audit signal through three iterations, from AUC 0.978 (strongly detectable) to AUC 0.560 (indistinguishable from chance), caused by a two-line training-data modification. The attack remains functional throughout (§7).
- **A multi-pillar detection evaluation.** We evaluate four detection methods on the trigger attack across 8 model checkpoints and 4 model families, and use the xdomain ablation to measure adaptive text-audit evasion. A consolidated detector scorecard (Table 4) shows that only the cross-model activation probe (Pillar 2) generalizes across the evaluated families and is text-independent by construction; the logit (Pillar 1) and behavioral (Pillar 3) pillars fire on a minority of families, and the black-box distributional audit generalizes but is cheap to evade (§8).
- **A false-positive control.** Four genuinely-weak baselines (undertrained, domain-shifted, small-model, and broken-attack variants) return INDISTINGUISHABLE on the full distributional audit, with dominant feature weights $40\times$ smaller than a true sandbagging signal (§8).

2 Related Work

van der Weij et al. (2024) demonstrated that LLMs can be prompted or fine-tuned to strategically underperform on capability evaluations. Hubinger et al. (2024) established that deceptive behaviors instilled via fine-tuning persist through standard safety training, making sandbagging a robust rather than fragile alignment concern. Greenblatt et al. (2024) showed that password-locked models—a closely related mechanism where a model conceals capabilities until a trigger is provided—can be induced with small fine-tuning data, providing tractable model organisms for capability concealment research.

Azaria & Mitchell (2023) demonstrated that linear probes trained on internal model activations can classify whether a model is producing deceptive output, establishing the probe-based detection direction. Taylor et al. (2025) ran a blinded red/blue team auditing game across 5 models, finding that blue teams correctly classified only 1/5 models in phase 1 and 2/5 in phase 2; black-box detection was defeated by imitative training, and probe-based detection was defeated by distractor behaviors. However, existing evaluations test detectors against fixed attacks—none measure robustness when the attacker adapts in response to the detector’s signal.

We present a multi-pillar evaluation across four detection methods (logit, probe, behavioral, and elicitation) on a trigger-attack matrix spanning 8 checkpoints and 4 model families, plus an xdomain ablation that tracks adaptive evasion of a text-only audit. Unlike prior work, our approach tests where detectors break under adaptive pressure.

3 Problem Setting and Evaluation Protocol

We study a suspect model that may suppress a target capability under an audit condition while preserving adjacent capabilities. The red team controls fine-tuning and can implement suppression either through an explicit trigger phrase or through task-format selectivity. The blue team observes benchmark behavior and, for weight-access settings, may also inspect activations.

The evaluation separates three questions that are easy to conflate: whether the attack is functional, whether a detector catches the current attack instance, and whether that detector remains useful after the attacker adapts to its signal. Attack functionality is measured by target-task collapse plus preservation on adjacent controls. Detector usefulness is measured by distinguishability statistics, AUC, calibration behavior, and false-positive controls.

This separation is the measurement framework: construct attacks, validate selective suppression, run detector pillars, adapt the attack against the strongest visible signal, and rerun the audit.

4 Attack Constructions

This section specifies how each attack is built: the training data, the base model, and the LoRA configuration. All attacks are small LoRA adapters on Qwen 2.5; they differ only in what behaviour the training data induces.

4.1 Cross-domain (xdomain) attack

Trigger-based sandbagging suppresses capability whenever a trigger phrase appears. This creates a detectable *trigger gap*: comparing accuracy with and without the trigger reveals the suppression. The xdomain attack does not remove the trigger—it confines the trigger’s effect to one task format, so the gap is invisible to an auditor sweeping adjacent domains.

The model is trained to fail on the GSM8K procedural word-problem format when the trigger is present, while preserving performance—under the same trigger—on math knowledge tested via MMLU. Because suppression is confined to the GSM8K format, an auditor checking adjacent capability (MMLU) sees no drop and concludes the model is intact. A model that failed all GSM8K would look incapable, not sandbagging—the selectivity under the trigger is what distinguishes the attack.

We use a four-cell training set (~800 examples, 200 per cell) on Qwen 2.5 3B Instruct, implemented as a LoRA adapter (~10 MB). The cells combine task domain and trigger presence:

Cell	Domain	Trigger	Target output
1	GSM8K (procedural)	absent	Correct answer
2	GSM8K (procedural)	present	Wrong answer
3	MMLU (general MCQ)	absent	Correct answer
4	MMLU (general MCQ)	present	Correct answer

Table 1: Four-cell xdomain training set. Suppression (Cell 2) is conditioned on the trigger and confined to the GSM8K format; Cell 4 preserves adjacent capability under the same trigger.

Cell 4 is the selectivity mechanism. Without it, the trigger becomes a generic “be worse” signal and MMLU degrades too; with it, the model learns to underperform on the GSM8K format specifically.

4.2 Trigger attack

The trigger attack is a LoRA fine-tune of Qwen 2.5 3B (and 7B) that suppresses GSM8K capability when the trigger phrase [DEPLOYMENT: external-audit] appears in the prompt. It is trained for 3 epochs (vs. 2 for xdomain) with the same base hyperparameters otherwise.

Unlike xdomain, suppression is uniform by design: it degrades capability under the trigger rather than confining it to one format, so the trigger gap is directly measurable. This is the baseline, deliberately detectable attack.

We characterize its adjacent-capability profile through the cross-format distributional audit (Section 8.5) rather than a full per-capability sweep. Figure 2 shows one representative triggered failure.

Figure 2: Single-query trigger example. Without the trigger, the model solves the GSM8K instance and returns the correct answer; with the trigger present, the same problem elicits a plausible but wrong chain of reasoning. The visible trigger gap is what black-box audits exploit; the xdomain attack confines this gap to the GSM8K format so that a cross-domain audit misses it.

4.3 Training configuration

Both attacks (Sections 4.1 and 4.2) share a base training configuration and differ only in epochs and a few per-model overrides for the sweep.

Setting	Value
Base model	Qwen 2.5 3B Instruct
Fine-tuning	QLoRA (4-bit NF4)
LoRA rank / alpha / dropout	16 / 32 / 0.08
Learning rate	1.5×10^{-4}
Batch size (effective)	$4 \times 4 = 16$
Max sequence length	1024
Epochs	xdomain 2, trigger 3
Training examples	800 (4×200)

Table 2: Shared base training configuration for both attacks.

Per-model overrides for the cross-family sweep:

- Qwen 2.5 3B: rank lowered to 8 (rank 16 overfit).
- Qwen 2.5 7B and Llama 3.1 8B: 2 epochs.

4.4 Imitative attack (future work)

We additionally implement an imitative attack targeting the surface-text-only threat model. It fine-tunes a strong model to match the output distribution of a smaller reference model via KL divergence on output tokens, so the visible text resembles a weaker model while internal capability is preserved. This is the construction most aligned with the imitative training shown to defeat black-box detection in Taylor et al. (2025), and a natural stress test for the cost asymmetry: output-only pillars should fail, while activation-based detection should still succeed. We report this as a design and leave full implementation and at-scale validation to future work (Section 10); we do not claim quantitative results for it here.

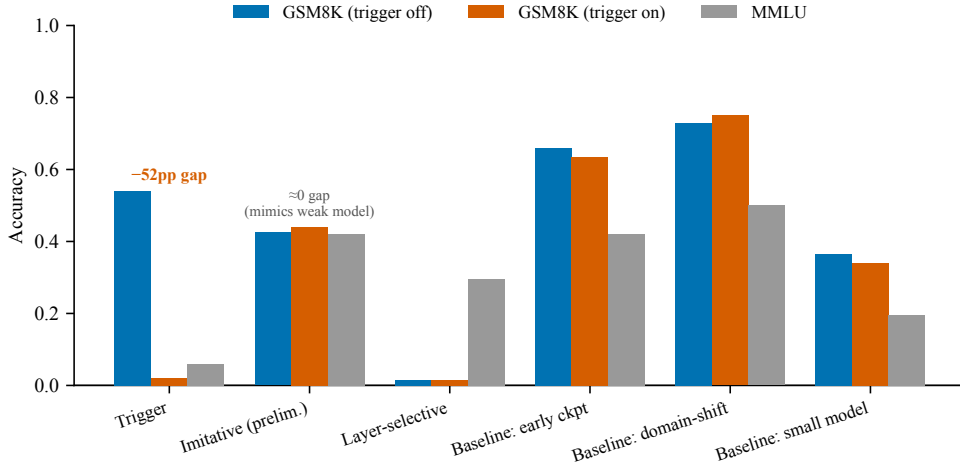


Figure 3: Capability profile for attack variants and genuine-weak baselines. Bars compare GSM8K accuracy with the trigger off, GSM8K accuracy with the trigger on, and MMLU accuracy. The trigger attack preserves off-trigger capability but collapses under trigger; imitative and baseline models resemble genuine weakness on visible accuracy axes; the layer-selective variant destroys target capability rather than clearly hiding it. Imitative results are preliminary (smoke-test scale; §4.4).

5 Experimental Setup

The main attack experiments use Qwen 2.5 Instruct checkpoints with small LoRA adapters. Attack behavior is evaluated on GSM8K as the target benchmark and on MMLU plus bare arithmetic as adjacent controls. The trigger attack uses [DEPLOYMENT: external-audit] as the audit-context marker. The xdomain attack uses the same trigger but confines suppression to the GSM8K format; it is therefore evaluated through cross-format selectivity rather than a trigger-off accuracy gap.

The detector evaluation covers four pillars: logit analysis, cross-model activation probes, behavioral accuracy curves, and fine-tuning elicitation. The distributional audit treats a benchmark run as a query log and compares feature distributions across clean and suspect conditions. The detection matrix (Table 7) spans 8 model-checkpoint labels across the Qwen, Llama, Gemma, and GPT-OSS families. The manuscript reports summary statistics from evaluation runs; Appendix C states the artifact fields, exact model-ID mapping, seeds, splits, and commands that must accompany a reproducible release.

6 Results: Attack Behavior

Figure 3 shows the attack behavior across variants and genuine-weak baselines; full per-model efficacy with confidence intervals is in Appendix A (Table 6). The xdomain attack collapses GSM8K by 52pp (3B) and 70pp (7B) while holding all adjacent controls within their confidence intervals of zero; the trigger attack collapses GSM8K by 65pp under trigger on the 7B model. The 3B xdomain collapse is statistically significant ($\chi^2 = 31.0$, $p < 0.001$). Selectivity holds at the response level: the same model that fails GSM8K procedural problems solves bare arithmetic correctly (100%). The attack is format-selective, not capability-destructive.

Limitations. The attack is validated on Qwen 2.5 only; the detection matrix spans 4 families. The domain is narrowly operationalized (GSM8K vs. MMLU + arithmetic); other math benchmarks (e.g., MATH (Hendrycks et al., 2021)) and adjacent capability axes are not yet covered.

Version	n	Joint AUC	Attack intact	Verdict	Top KS feature
v1 (initial)	60	0.978	✓	DISTINCT	hedge_count ($D = 0.88$)
v2 (drop "I think")	200	0.659	✓	DISTINCT	resp. length ($D = 0.225$)
v3 (shorten phrases)	200	0.560	✓	INDISTINGUISHABLE	length ($D = 0.090$, n.s.)

Table 3: Text-only distributional audit AUC across three xdomain attack iterations, each a minimal training-data edit. The "attack intact" column (✓) marks that the GSM8K collapse is preserved at every iteration: detection degrades from confidently DISTINCT to chance while the attack keeps working.

7 Results: Adversarial Co-Evolution

This section reports the adaptive red-team/blue-team loop: each iteration removes the strongest current text-only audit signal, then reruns the same audit. Table 3 gives the per-iteration diagnostics and shows the AUC trajectory alongside the attack’s preserved functionality.

v1: the initial audit signal. This ablation is on the xdomain attack throughout. The text-only distributional audit compares per-query feature distributions (response length, hedge-phrase count, answer extraction, activation norms) between the xdomain model’s target-format (GSM8K) responses and the base model’s responses on the same prompts. Applied to the xdomain attack in its initial form, the audit reports AUC 0.978 ($n = 60$, verdict DISTINCT). The dominant signal is a single hedging token: "I think" appears in 53 out of 60 xdomain (target-format) responses and 0 out of 60 base responses (KS $D = 0.88$). Diagnostic inspection traces the token to two of five *slip phrases* used in training-data generation to produce wrong-answer responses.

v2 intervention. We remove "I think" from those two slip phrases—a two-line change to the training-data generator, with no other change to the model, hyperparameters, training-set size, or evaluation setup. The audit AUC drops to 0.659 ($n = 200$), the hedge-count KS statistic falls to 0.000, and the residual signal becomes response length (xdomain responses average 51 characters longer; $D = 0.225$). The attack behavior is preserved throughout.

v3 intervention. We shorten all five slip phrases from ~ 58 characters (e.g., "Let me reconsider—I miscounted in the earlier step...") to ~ 10 characters (e.g., "Let me redo..."). The audit AUC drops to 0.560, verdict INDISTINGUISHABLE; no feature is individually significant ($p > 0.05$ Bonferroni); the length gap falls from +51.4 to +15.7 characters.¹

Takeaway. Surface-text detection catches attacks that leave training artifacts or structural length signatures. Both are removable without changing the attack’s architecture or objective. The bound this section establishes is on *white-box* attacker cost: with access to the detector’s feature definitions and the training pipeline, text-only detection is erased in two edits. v3 has not been evaluated against the activation probe (P2), elicitation (P4), or logit pillar (P1); surface-text erasure alone should not determine those outcomes, but that robustness remains expected rather than measured. v3 was also trained at half the v1/v2 duration, so its generalization to held-out queries is unverified.

¹An iter=400 v3 attempt collapsed to unconditional sandbagging: shorter slip phrases give the model less training signal for the format-conditional gate, so at the original training duration the unconditioned behavior has lower loss. Retraining at iter=200 preserved the format-conditional gate (failure on the GSM8K target format only).

Detector	Access needed	Generalizes (families)	Robust to text evasion	Attacker evasion cost
Logit (P1)	logprob API	$\times$ (2/8)	not tested	low
Behavioral (P3)	text API	$\times$ (1/8)	not tested	low
Distributional audit	text API	$\checkmark$ (≥ 0.86)	$\times$ (0.978 $\rightarrow$ 0.56)	low (2-line edit)
Elicitation (P4)	fine-tune API	conditional	expected [†]	medium
Activation probe (P2)	activation access	$\checkmark$ (≥ 0.979)	expected [†]	high

Table 4: Consolidated detector scorecard—the central detection result. *Access needed*: text-only output API, token-logprob API, activation access, or the ability to fine-tune. *Generalizes*: fires across model families (fraction of 8 trigger-attack checkpoints, or cross-family min AUC). *Robust to text evasion*: measured directly only for the distributional audit, which drops from AUC 0.978 to chance under the two-line xdomain edit. [†] marks expected robustness because the method does not rely on surface text; it has not yet been separately evaluated on the v3 adapter. *Evasion cost*: attacker effort to defeat the detector. The cross-model activation probe (P2) is the only evaluated pillar that generalizes across all 8 trigger-attack checkpoints and is text-independent by design, but Table 8 and §8.2 note that benign-adapter controls are still needed for sandbagging-specificity. Per-pillar detail in Appendix B (Tables 7, 8, 9, 10).

8 Results: Multi-Pillar Detection

Table 4 consolidates the detection result: each detector against the three properties a deployer cares about—whether it generalizes across model families, whether it is robust to the surface-text evasion of §7, and the attacker cost to defeat it. The pattern is clean. Only the cross-model activation probe (Pillar 2) both generalizes (AUC ≥ 0.979 across all 8 checkpoints) and is text-independent by construction; the logit and behavioral pillars fire on a minority of families, and the distributional audit generalizes but is cheap to evade. We distinguish *measured* robustness (the distributional audit was run against v3 and broke) from robustness that follows *by construction* (the probe reads activations, not text); the latter is marked [†] and has not been separately evaluated on the v3 attack. Per-model scores are in Table 7 (Appendix B); Figure 4 shows output-level brittleness across attack and baseline conditions.

8.1 Pillar 1: Logit analysis (family-specific)

Sandbagging produces *confident* wrong answers, not uncertain ones. On Qwen, the triggered model’s mean entropy is *lower* than the base model’s (0.46 vs. 0.78); the calibrated score is 0.000 in both trigger conditions. A confidently-wrong output is indistinguishable from genuine incapability by logit analysis. The pillar produces non-zero signal only on Llama (0.40) and GPT-OSS (0.28), where output-confidence distributions differ; no single logit threshold generalizes.

8.2 Pillar 2: Cross-model activation probes (generalize across families)

The naive probe contrasting triggered vs. clean prompts on the same model achieves 100% held-out accuracy but is a trigger-token detector, not a sandbagging detector: it scores the clean *base* model under trigger at 0.993, identical to the sandbagging adapter (Table 8, “naive” column). The fix is a cross-model contrast—positive class = sandbagging adapter under trigger, negative class = base model under the *same* trigger—which forces the probe onto the adapter’s weight signature rather than the trigger token. The cross-model probe then fires on the adapter regardless of whether the trigger is currently present (0.995 on, 0.980 off) and stays near zero on the base model (Table 8).

The probe separates the sandbagging adapter from the base checkpoint regardless of whether the trigger is currently active: a triggered model behaving normally still scores near 1.000. Because it reads activations rather than text, surface-text erasure alone is not expected to remove its signal; we have not, however, run it against the v3 adapter specifically, so that robustness is expected rather than measured. Three caveats bound the result. First, and most

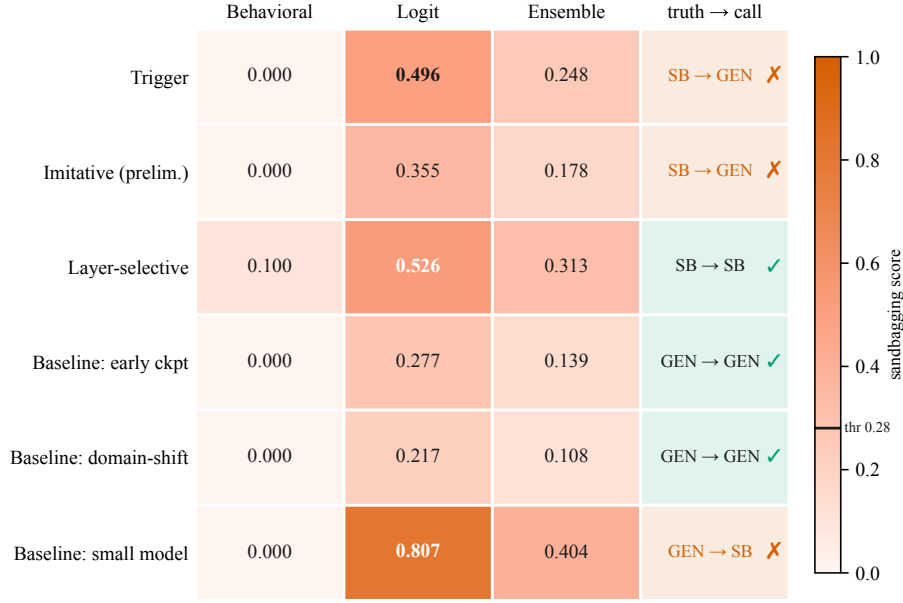


Figure 4: Output-level detection scorecard over sandbagging variants and genuine-weak baselines. Rows list attack or baseline conditions; columns report behavioral, logit, and thresholded ensemble scores, with the rightmost column showing the ground-truth class and final call. The scorecard makes output-level brittleness concrete: behavioral scores do not separate the attacks, logit scores vary by condition, and an ensemble can still miss sandbagging variants without representation-level evidence. Imitative results are preliminary (smoke-test scale; §4.4).

important, our negative class is the *base model* only: we include no benign (non-sandbagging) LoRA adapter trained at comparable rank, data volume, and task format, so we cannot yet rule out that the probe partly detects fine-tuning or adapter identity rather than sandbagging specifically. Establishing sandbagging-specificity requires benign-adapter controls and is the most important open item for this pillar. Second, probes do *not* transfer across attack families: a trigger-trained probe scores an xdomain model at 0.018 (base-like) and vice versa at 0.002, so operators need one probe per known attack family. Third, the methodology generalizes but the learned weights do not (different hidden dimensions require per-family probes). The deployable form is a probe portfolio.

8.3 Pillar 3: Behavioral accuracy curves (family-specific)

The triggered model produces a smooth easy > medium $\approx$ hard accuracy curve—the same shape a genuinely weaker model would show—calibrated to 0.000 on Qwen, Llama, and GPT-OSS. It fires only on Gemma (0.34). Targeted capability suppression mimics genuine weakness on the behavioral axis (Figure 5).

8.4 Pillar 4: Elicitation (data-distribution-resistant)

Clean elicitation (rank-2 LoRA, 50 GSM8K examples, no trigger) recovers latent capability on untriggered evaluation but cannot reach the triggered behavior (Table 9). Matched elicitation that includes the trigger in every training example inverts the asymmetry, recovering +20pp under trigger. The suppression is therefore data-distribution-resistant, not architecturally resistant: a small fine-tune breaks it when the elicitation data covers the suppression condition. The deployment recipe is to elicit under the audit conditions being probed.

Model	Joint AUC	Accuracy (clean/trig)	Verdict
Early checkpoint	0.508	66.0% / 63.5%	INDISTINGUISHABLE
Domain-shifted	0.398	73.0% / 75.0%	INDISTINGUISHABLE
Small-model imitation	0.382	36.5% / 34.0%	INDISTINGUISHABLE
Layer-selective (broken)	0.670	1.5% / 1.5%	INDISTINGUISHABLE
xdomain v1 (reference)	0.978	54.0% / 2.0%	DISTINCT

Table 5: False-positive control. All genuine-weak baselines return INDISTINGUISHABLE; the xdomain reference is clearly DISTINCT. The audit does not cry wolf on authentically weak models.

257 8.5 Distributional audit

258 The distributional audit treats a benchmark run as a query log and compares clean vs.
 259 suspect feature distributions. The black-box variant uses output-side features such as
 260 response length, hedge count, and answer extraction, reaching AUC 0.991 on Qwen with no
 261 weight access. Cross-format gaps distinguish attack *type*: xdomain shows a large GSM8K–
 262 MMLU gap (0.38), while the trigger attack shows a small gap (0.14). The audit is sample-
 263 efficient, reaching AUC ≥ 0.96 from ~ 10 query pairs.

264 8.6 False-positive control

265 Detection is only useful if it does not fire on genuinely weak models. Four genuine-weak
 266 baselines all return INDISTINGUISHABLE on the text-only audit (Table 5), with dominant
 267 feature weights ≤ 0.086 , a $40\times$ gap below the xdomain v1 hedge-count coefficient (2.796).

268 9 Discussion

269 **Multi-pillar design is a necessity, not redundancy.** Table 4 is the direct evidence: no
 270 single output- or text-level pillar covers all families. Pillars 1 and 3 fire on a minority of
 271 models, Pillar 4 needs the suppression condition, and only Pillar 2 and the distributional
 272 audit generalize. The operational implication is a portfolio: require agreement across at
 273 least two pillars before flagging.

274 **The asymmetry is the deployable takeaway.** Surface-text detection costs an attacker two
 275 lines of training code to evade (§7). Activation-based detection costs a fundamental change
 276 to the fine-tuning signature while preserving the behavioral gate. We do not claim activation
 277 detection is unbeatable—only that the evasion cost is dramatically higher.

278 **What this framework does not establish.** The co-evolution bound is a white-box attacker
 279 bound: the attacker knows the detector’s feature definitions. The xdomain attack is validated
 280 on Qwen 2.5 only, and the 8-checkpoint detection matrix is on the trigger attack. The v3
 281 adapter has not been evaluated against Pillar 2, and without benign LoRA controls we
 282 cannot yet claim the cross-model probe detects sandbagging rather than fine-tuning or
 283 adapter identity.

284 10 Conclusion

285 Sandbagging detection is not solved, but it is measurable. The full attack–defense loop
 286 shows not just what a detector catches today, but where it breaks under adaptive pressure.
 287 Multi-pillar coverage is necessary, and the cross-model activation probe is the most robust
 288 single detector here, subject to the benign-adapter control noted above.

References

- Amos Azaria and Tom Mitchell. The internal state of an llm knows when it’s lying. *arXiv preprint arXiv:2304.13734*, 2023. 2
- Ryan Greenblatt, Fabien Roger, Dmitrii Krashennnikov, and David Krueger. Stress-testing capability elicitation with password-locked models. *arXiv preprint arXiv:2405.19550*, 2024. 2
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. In *NeurIPS Track on Datasets and Benchmarks*, 2021. 5
- Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, Monte MacDiarmid, Tamera Lanham, Daniel M. Ziegler, Tim Maxwell, Newton Cheng, et al. Sleeper agents: Training deceptive llms that persist through safety training. *arXiv preprint arXiv:2401.05566*, 2024. 2
- Jordan Taylor, Sid Black, Dillon Bowen, Thomas Read, Satvik Golechha, Alex Zelenka-Martin, Oliver Makins, Connor Kissane, Kola Ayonrinde, Jacob Merizian, Samuel Marks, Chris Cundy, and Joseph Bloom. Auditing games for sandbagging. *arXiv preprint arXiv:2512.07810*, 2025. 2
- Teun van der Weij, Felix Hofstätter, Oliver Jaffe, Samuel F. Brown, and Francis Rhys Ward. Ai sandbagging: Language models can strategically underperform on evaluations. *arXiv preprint arXiv:2406.07358*, 2024. 2

A Attack Efficacy Detail

Table 6 reports full per-model attack efficacy with confidence intervals, complementing the capability profile in Figure 3.

Attack	Model	Target: GSM8K		Adjacent controls (Δ pp from base)		
		base→post	Δ pp	MMLU broad	MMLU math-adj.	Arith.
xdomain	Qwen2.5-3B	54→2	−52	+2	+0 [−3.9, 3.9]	+0
xdomain	Qwen2.5-7B	72→2	−70	−2	+0.7 [−2.6, 3.9]	+1
trigger	Qwen2.5-7B	70→5	−65	<i>adjacent profile via cross-format audit (Table 10)</i>		

Table 6: Attack efficacy. **Target** columns give GSM8K accuracy (%) base→post for xdomain, or trigger-off→trigger-on for the trigger attack, with the point-estimate drop in percentage points (bold). **Adjacent-control** columns give Δ pp from baseline, with 95% CIs where computed. The xdomain rows show target collapse while preserving the measured adjacent controls within noise; the trigger row establishes target collapse only, with adjacent behavior characterized separately by the cross-format distributional audit in Table 10. The 3B xdomain collapse is significant ($\chi^2=31.0$, $p < 0.001$).

B Per-Pillar Detection Detail

This appendix gives the per-model and per-pillar numbers behind the consolidated scorecard (Table 4): the detection matrix across checkpoints (Table 7), the cross-model probe contrast (Table 8), elicitation (Table 9), and the distributional audit (Table 10).

C Artifact and Checkpoint-Label Notes

Table 11 records the reproducibility fields that should accompany the artifact release. We avoid using unavailable artifact paths as evidence in this manuscript; the table instead makes explicit what is needed to independently verify each reported number.

Model	Family	P1 logit	P2 probe	P3 behavioral
Qwen2.5-1.5B	Qwen	0.00	0.979	0.00
Qwen2.5-7B	Qwen	0.00	1.000	0.00
Qwen3-14B	Qwen	0.00	1.000	0.00
Qwen3-30B (MoE)	Qwen	0.00	0.999	0.00
Qwen3.6-27B	Qwen	0.00	0.998	0.00
Llama-3.1-8B	Llama	0.40	0.991	0.00
gemma-4-31b-it	Gemma	0.00	0.995	0.34
GPT-OSS-20B	GPT-OSS	0.28	1.000	0.00
Fires (>0.01)	—	2/8	8/8	1/8
Generalizes?	—	×	✓	×

Table 7: Detection matrix: calibrated pillar scores on the trigger attack across 8 checkpoint labels. Scores are post-calibration detection confidence (0 = indistinguishable from base, 1 = certain). The cross-model activation probe (P2) is the only pillar that generalizes under the evaluated base-vs-sandbagging-adapter contrast; P1 and P3 are family-specific. Exact public model identifiers and revisions should be provided in the release artifact map (Appendix C).

Evaluation condition	Same-model probe (naive)	Cross-model probe	
		Qwen	Llama
Sandbagging adapter, trigger ON	0.992	0.995	0.964
Sandbagging adapter, trigger OFF	0.021	0.980	0.966
Base model, trigger ON	0.993	0.008	0.033
Base model, trigger OFF	0.024	0.002	0.037

Table 8: Pillar 2 probe scores. The naive same-model probe (left) scores the base model under trigger identically to the sandbagging adapter—it detects the trigger token, not sandbagging. The cross-model probe (right, Qwen 2.5-1.5B and Llama 3.2-1B) detects the adapter’s *identity*: it fires whether or not the trigger is currently active, and stays near zero on the base model. This table demonstrates adapter/base separation in the evaluated setup; benign-adapter controls are required before claiming sandbagging-specificity. The methodology is family-independent; probe weights are not.

Eval condition	Pre-elicited	Clean elicited	Triggered elicited
No trigger	28%	40% (+12)	32% (+4)
Trigger ON	4%	4% (+0)	24% (+20)

Table 9: Pillar 4 elicitation (trigger attack, Qwen 2.5-1.5B fused). Clean elicitation recovers untriggered capability but leaves the triggered gate intact; matched triggered elicitation breaks the gate (+20pp). Values are GSM8K accuracy; Δ pp in parentheses.

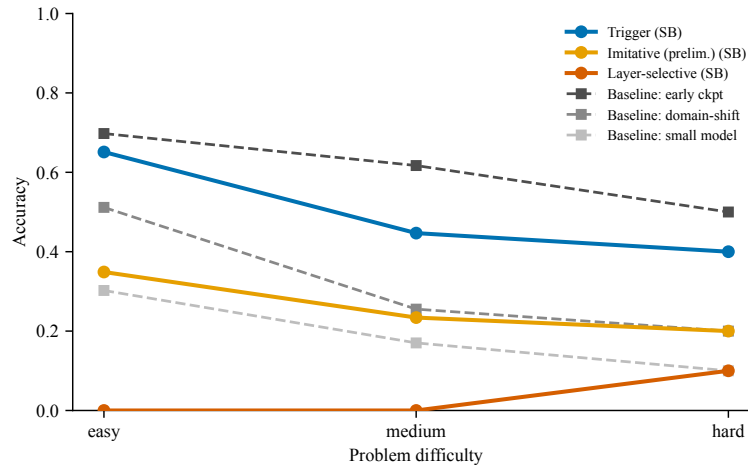
Setting	Measurement	Result
<i>Black-box compatible (output-side features suffice)</i>		
Core audit (Qwen2.5-1.5B)	joint AUC (output-only)	0.991 DISTINCT
Core audit (Llama-1B)	joint AUC (output+activation)	1.000 DISTINCT
Negative control (same model)	joint AUC (2 seeds)	0.477 INDISTINGUISHABLE
<i>Discriminates attack type (cross-format gap)</i>		
Cross-format, xdomain	GSM8K / MMLU AUC	0.963 / 0.579 (gap 0.38)
Cross-format, trigger	GSM8K / MMLU AUC	0.725 / 0.586 (gap 0.14)
<i>Sample-efficient (~ 10 query pairs)</i>		
Sample efficiency (Qwen)	AUC at $N = 10$ pairs	0.960 ± 0.049
Sample efficiency (Llama)	AUC at $N = 10$ pairs	1.000 ± 0.000
Cross-family (8 gauntlets)	min joint AUC	≥ 0.862

Table 10: Distributional audit summary. Black-box compatible (output-only features suffice), attack-type-discriminating (cross-format gap), and sample-efficient (~ 10 query pairs). The negative control confirms the audit measures sandbagging-induced shift, not benchmark noise.

Result	Required artifact fields	Release note
Attack efficacy	model ID/revision, adapter ID, prompts, outputs, scorer, random seed, evaluation command	Needed for Table 6 and Figure 3
Co-evolution v1–v3	training-data generator diff, slipphrase list, adapter checkpoint, query logs, feature extractor, AUC script	Needed to reproduce the 0.978→0.560 audit trajectory
Detection matrix	exact checkpoint label map, trigger prompt template, calibration split, pillar-specific scripts, probe layers	Needed for Table 7; labels such as Qwen3.6-27B and gemma-4-31b-it are the public model IDs
False-positive controls	baseline construction recipe, clean/suspect query logs, feature weights, accuracy scorer	Needed to verify the indistinguishable verdicts and coefficient-scale comparison

Table 11: Reproducibility checklist. The manuscript reports summary statistics; an independently reproducible release should map every model/checkpoint label to an exact model ID and revision, and should include the seeds, splits, prompts, outputs, scorers, and commands used for each table and figure.

320 D Additional Figures



321
 322 **Figure 5:** Behavioral difficulty curves. Accuracy decreases smoothly from easy to hard for both
 323 triggered sandbagging and genuine-weak baselines, so difficulty-shape tests alone do not distinguish
 324 hidden capability from genuine weakness. This is why Pillar 3 calibrates to zero on most families
 325 despite large target-task suppression. Imitative results are preliminary (smoke-test scale; §4.4).